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Understanding the ACL in order to train and condition

The evaluation begins by looking for successes in order to determine where to start our prevention/rehabilitation program . . . How will the training and conditioning program be blended to prepare the entire body for all types of functional activities?

Gait Analysis with shoes on and shoes off

Looking at the feet and their relationship to the knee
 Looking for the ability of the feet to load and unload in all three planes
 Single leg balance trunk rotation looking for collapsing of the arch with calcaneal eversion.
 Single leg balance with knee flexion excursion looking for ankle dorsiflexion
 Single leg balance with knee flexion excursion with trunk rotation with knee abduction drive and knee adduction drive
 Looking at the ability to load and unload in all three planes of motion
 Quick biomechanical scan of the feet looking at position, angulations and motions
 Lower leg to calcaneus relationships
 Calcaneus to forefoot relationships
 Callus patterns

Looking at the hip and it's relationship to the knee and foot

Single leg balance with opposite leg anterior to posterior reach (sagittal plane drive)
 Single leg balance with opposite leg lateral to medial reach (frontal plane drive)
 Single leg balance with opposite leg right rotational to left rotational reach (transverse plane drive)
 Single leg balance with bilateral U. E. anterior reach at waist height to overhead posterior reach (sagittal plane drive)
 Single leg balance with bilateral U.E. overhead lateral to overhead medial reach (frontal plane drive)
 Single leg balance with bilateral U.E. right rotational reach to left rotational reach at shoulder height (transverse plane drive)

Looking at preloaded rotational excursion tests

Single leg balance, toe touch rotational reach to facilitate internal rotation, opposite arm same side overhead rotational reach
 Single leg balance, toe touch rotational reach to facilitate external rotation, opposite arm same side overhead rotational reach



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Step down with return testing

Anterior, posterior and medial step down and return test
Medial rotational, lateral rotational step down and return test

Description of tweak progressions

Knee flexion excursion tests with feeding in and feeding out

Hip flexion (feed in) with bilateral U.E. anterior reach at knee height
Hip adduction (feed in) with bilateral U.E. overhead medial overhead reach
Hip internal rotation (feed in) with bilateral U.E. same side rotational reach at shoulder height
Triplane (feed in) with opposite arm anterior lateral rotational reach at knee height
Hip extension (feed out) with bilateral overhead posterior reach
Hip abduction (feed out) with bilateral U.E. overhead lateral reach
Hip external rotation (feed out) with bilateral U.E. opposite side rotational reach at shoulder height
Triplane (feed out) with bilateral U.E. overhead posterior lateral rotational reach

Description of bottom up tweaks in the sagittal plane, frontal plane and transverse plane